

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V) Artificial
Intelligence (AI)

Practical 1

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1) Design a Simple Rational Agent by defining PEAS for Vacuum Cleaner Environment and Autonomous Taxi. Classify environments (fully/partially observable, deterministic/stochastic, episodic/sequential). Implement a simple table-driven agent in Python.

Vacuum Cleaner Environment Code :-

```
table = {  
    ('A', 'Dirty'): 'Suck',  
    ('A', 'Clean'): 'Right',  
    ('B', 'Dirty'): 'Suck',  
    ('B', 'Clean'): 'Left',  
    ('C', 'Dirty'): 'Suck',  
    ('C', 'Clean'): 'Backward',  
    ('D', 'Dirty'): 'Suck',  
    ('D', 'Clean'): 'Forward'  
}  
  
sequence_table = {  
    (('A', 'Dirty'), ('A', 'Clean')): 'Right',  
    (('A', 'Clean'), ('B', 'Dirty')): 'Suck',  
    (('B', 'Dirty'), ('B', 'Clean')): 'Left',
```

```

(('B', 'Clean'), ('A', 'Dirty')): 'Suck',
(('C', 'Dirty'), ('C', 'Clean')): 'Backward',
(('C', 'Clean'), ('A', 'Dirty')): 'Suck',
(('D', 'Dirty'), ('D', 'Clean')): 'Forward',
(('C', 'Clean'), ('C', 'Dirty')): 'Suck'
}
percepts = []
def table_driven_agent(percept):
    percepts.append(percept)
    if tuple(percepts) in sequence_table:
        return sequence_table[tuple(percepts)]
    return table.get(percept, "NoOp")
print("Vacuum Cleaner Table-Driven Agent")
n = int(input("Enter number of percepts: "))
for i in range(n):
    print(f"\nPercept {i+1}")
    location = input("Enter Location (A/B/C/D): ").upper()
    status = input("Enter Status (Clean/Dirty): ").capitalize()

    percept = (location, status)
    action = table_driven_agent(percept)
    print("Percept History :", percepts)
    print("Agent Action :", action)
    print("Saqlain Khan SO13")

```

Output :-

```

Percept 1
Percept History : [('A', 'Dirty')]
Agent Action    : Suck
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Percept 2
Percept History : [('A', 'Dirty'), ('B', 'Clean')]
Agent Action    : Left
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Percept 3
Percept History : [('A', 'Dirty'), ('B', 'Clean'), ('C', 'Dirty')]
Agent Action    : Suck
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Percept 4
Percept History : [('A', 'Dirty'), ('B', 'Clean'), ('C', 'Dirty'), ('D', 'Clean')]
Agent Action    : Forward
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Percept 5
Percept History : [('A', 'Dirty'), ('B', 'Clean'), ('C', 'Dirty'), ('D', 'Clean'), ('A', '')]
Agent Action    : NoOp
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```

Autonomous Taxi Code :-

```
table = {
```

```
    ('Green', 'No', 'No'): 'Move Forward',
```

```
    ('Green', 'Yes', 'No'): 'Stop',
```

```
    ('Red', 'No', 'No'): 'Stop',
```

```
    ('Yellow', 'No', 'No'): 'Slow Down',
```

```
    ('Green', 'No', 'Yes'): 'Pick Up Passenger',
```

```
    ('Red', 'Yes', 'No'): 'Stop',
```

```
    ('Yellow', 'Yes', 'No'): 'Stop',
```

```
    ('Green', 'Yes', 'Yes'): 'Wait Until Road is Clear',
```

```
    ('Yellow', 'No', 'Yes'): 'Slow Down and Pick Up Passenger',
```

```
    ('Red', 'No', 'Yes'): 'Wait for Green Signal'
```

```
}
```

```
percepts = []
```

```
def taxi_agent(percept):
```

```
    percepts.append(percept)
```

```

    action = table.get(percept, "Wait")

    return action

print("Autonomous Taxi Table-Driven Agent")

n = int(input("Enter number of percepts: "))

for i in range(n):

    print(f"\nPercept {i+1}")

    signal = input("Traffic Signal (Green/Yellow/Red): ").capitalize()
    obstacle = input("Obstacle Present? (Yes/No): ").capitalize()
    passenger = input("Passenger Request? (Yes/No): ").capitalize()

    percept = (signal, obstacle, passenger)

    action = taxi_agent(percept)

    print("Current Percept:", percept)

    print("Agent Action:", action)

    print("Saqlain Khan SO13")

```

Output :-

```

Autonomous Taxi Table-Driven Agent

Percept 1
Current Percept: ('Green', 'No', 'No')
Agent Action: Move Forward
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Percept 2
Current Percept: ('Yellow', 'Yes', 'Yes')
Agent Action: Wait
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Percept 3
Current Percept: ('Red', 'No', 'Yes')
Agent Action: Wait for Green Signal
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Percept 4
Current Percept: ('Green', 'Yes', 'No')
Agent Action: Stop
Saqlain Khan SO13

```

Gaming Driven Table Code :-

```

table = {
    ('Yes', 'No', 'No'): 'Attack Enemy',
    ('Yes', 'Yes', 'No'): 'Use Health Kit',
    ('Yes', 'No', 'Yes'): 'Reload Weapon',
    ('No', 'No', 'No'): 'Explore Map',
    ('No', 'Yes', 'No'): 'Search for Health Pack',
    ('No', 'No', 'Yes'): 'Search for Ammo',
    ('Yes', 'Yes', 'Yes'): 'Retreat',
    ('No', 'Yes', 'Yes'): 'Find Safe Zone',
    ('No', 'No', 'Yes'): 'Collect Ammo Crate',
    ('Yes', 'No', 'No'): 'Eliminate Enemy'
}

percepts = []

def gaming_agent(percept):
    percepts.append(percept)
    action = table.get(percept, "Wait")
    return action

print("Gaming Table-Driven Agent")

n = int(input("Enter number of percepts: "))
for i in range(n):
    print(f"\nPercept {i+1}")
    enemy = input("Enemy Nearby? (Yes/No): ").capitalize()
    health = input("Health Low? (Yes/No): ").capitalize()
    ammo = input("Ammo Low? (Yes/No): ").capitalize()
    percept = (enemy, health, ammo)

```

```
action = gaming_agent(percept)
print("Current Percept:", percept)
print("Agent Action:", action)
print("Saqlain Khan SO13")
```

Output :-

```
Gaming Table-Driven Agent

Percept 1
Current Percept: ('Yes', 'Yes', 'No')
Agent Action: Use Health Kit
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Percept 2
Current Percept: ('No', 'No', 'Yes')
Agent Action: Collect Ammo Crate
Saqlain Khan SO13

Percept 3
Current Percept: ('No', 'No', 'No')
Agent Action: Explore Map
Saqlain Khan SO13
```